



SIMATS ENGINEERING

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Slot: C

Project Title: WASTE MANAGEMENT SYSTEM

Module: DICTIONARY OPERATIONS

Module Photographs : (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)

The screenshot displays the 'Smart Waste Collection and Recycling Management System' interface. It features two main panels. The left panel is a form for adding waste, with fields for 'Waste Type' (food waste), 'Quantity (kg)' (5), 'Category' (wet waste), and 'Recyclable (Yes/No)' (no). Below these fields are buttons for 'ADD WASTE', 'DISPLAY WASTE', 'SEARCH WASTE', 'UPDATE WASTE', 'DELETE WASTE', 'SAVE DATA', and 'LOAD DATA'. The right panel shows a 'Search Result' window with the following data: Waste type: food waste, Quantity: 5 kg, Category: wet waste, Recyclable: no. A 'Waste Records' window is also visible, listing waste entries with their types, quantities, categories, and recyclability status.

Waste Type	Quantity (kg)	Category	Recyclable (Yes/No)
food waste	5	wet waste	no

Waste Type	Quantity (kg)	Category	Recyclable (Yes/No)
Waste type: food waste	Quantity: 5 kg	Category: wet waste	Recyclable: no
Waste type: paper	Quantity: 10 kg	Category: dry waste	Recyclable: yes
Waste type: plastic	Quantity: 10 kg	Category: dry waste	Recyclable: yes
Waste type: paper	Quantity: 10 kg	Category: dry waste	Recyclable: yes
Waste type: plastic	Quantity: 5 kg	Category: dry waste	Recyclable: yes
Waste type: food waste	Quantity: 5 kg	Category: wet waste	Recyclable: no

Project Description : (here you write what you did in this project (contribution) including Model Description)

Individual Contribution – Module 1: Dictionary Operations

The first module of the **Smart Waste Collection and Recycling Management System** focuses on managing waste information using Python dictionaries. The main objective of this module is to provide an efficient way to store, access, modify, and remove waste records. I contributed to the design and implementation of the dictionary-based operations used in the project.

I implemented the waste data dictionary to store information about different types of waste. The waste type is used as the dictionary key, while the quantity, category, and recyclable status are stored as values inside another dictionary. This structure allows multiple details of each waste type to be stored in an organized manner.

I worked on the **Add Waste** operation, where the user enters the waste type, quantity, category, and recyclable status through the Tkinter input fields. The entered information is then stored in the dictionary as a key-value pair. I also implemented input validation to ensure that the waste type is entered before adding a record.

I contributed to the **Display Waste** operation by using the items () method of the dictionary. This operation accesses every waste type and its corresponding details and displays them to the user. The **Search Waste** operation was implemented using the in operator to check whether a particular waste type exists in the dictionary. If the record exists, its details are retrieved and displayed; otherwise, an appropriate error message is shown.

I also worked on the **Update Waste** operation. When an existing waste type is entered, its quantity, category, and recyclable status can be replaced with new values. The **Delete Waste** operation uses the del statement to remove an existing waste record from the dictionary.

Testing and debugging were an important part of my contribution to this module. I tested different waste types such as Plastic, Paper, and Food Waste and verified that adding, displaying, searching, updating, and deleting records worked correctly. I also checked error conditions such as empty input fields and searching for records that do not exist.

Through this module, I gained practical knowledge of Python dictionaries, key-value pairs, nested dictionaries, dictionary methods, conditional statements, loops, functions, input validation, and debugging. This module forms the main data management component of the waste management system.

Student Signature

Guide Signature